



Ant foraging with obstacles

The idea

- During ant foraging, ants do not proceed linearly: they are very likely to encounter obstacles along the way
- **Goal:** implement a system to simulate and describe the behavior of ants during foraging, when they encounter obstacles



Initial considerations

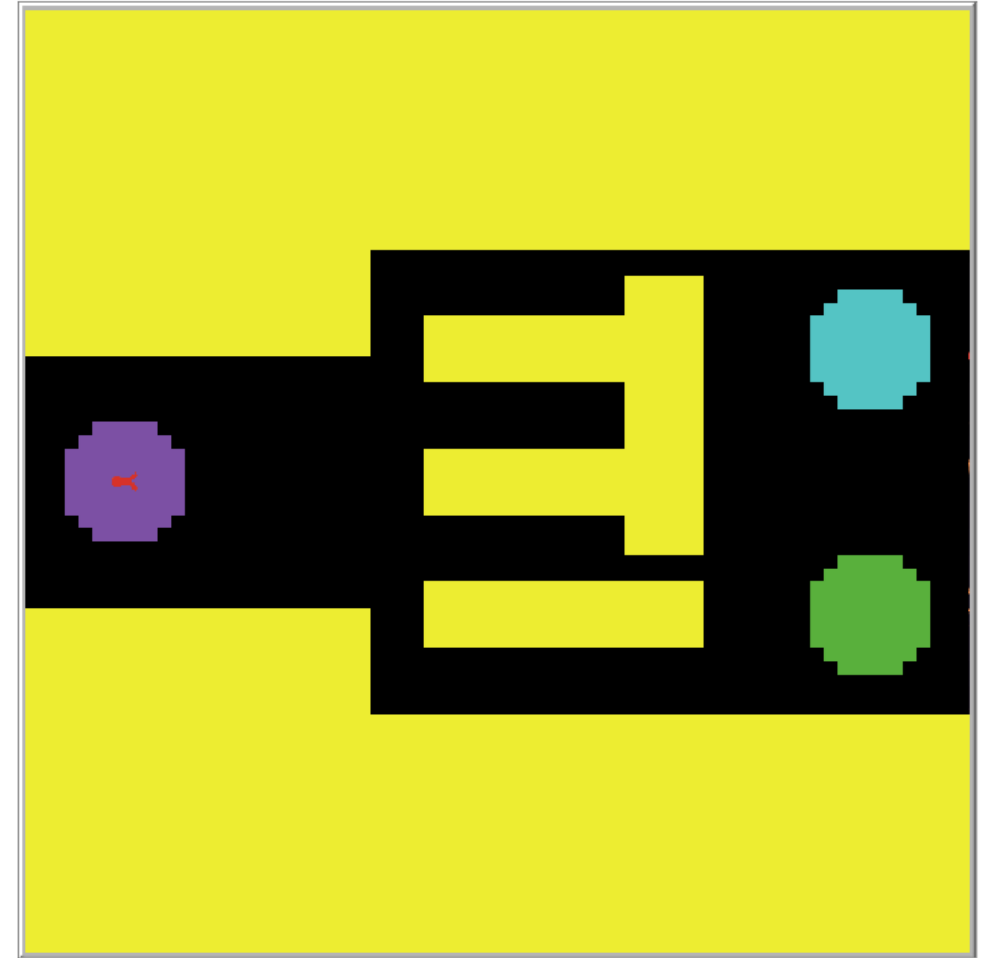
There are several aspects to take into consideration, as explained in the paper **[1]**:

- Avoid obstacles of any kind without crashing into them multiple times
- Wander randomly, in the general direction of any pheromones (scent markers that many insects generate) nearby
- If the ant is holding food, it releases the pheromone at a constant rate as it walks. In the simplest simulations, the ant continues to move randomly
- If the ant finds food and does not have any in its hand, it picks it up
- If the ant is in the nest and carrying food, it places it in and goes back searching

The model

The model is structured in this way:

- **Red:** ants
- **Cyan and green:** food sources
- **Purple:** nest
- **Yellow:** obstacles

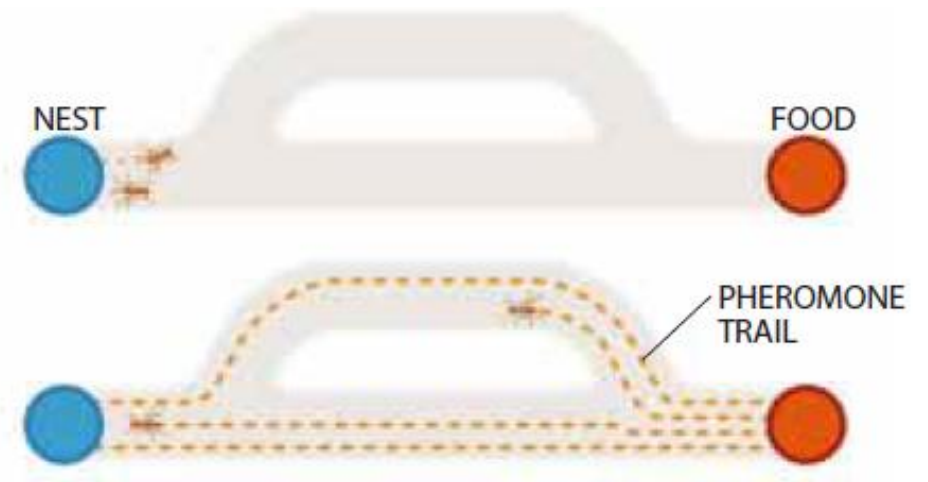


Obstacles structure

- The obstacles are arranged randomly, but in such a way as to generate well-defined and complex paths
- The model finds inspiration from the ants of the tropical forests of Mexico **[2]** that have to find a path that leads them to food by following trees' branches
- With the structure shown we simulate a behavior where all the paths are different and more than one leads to food

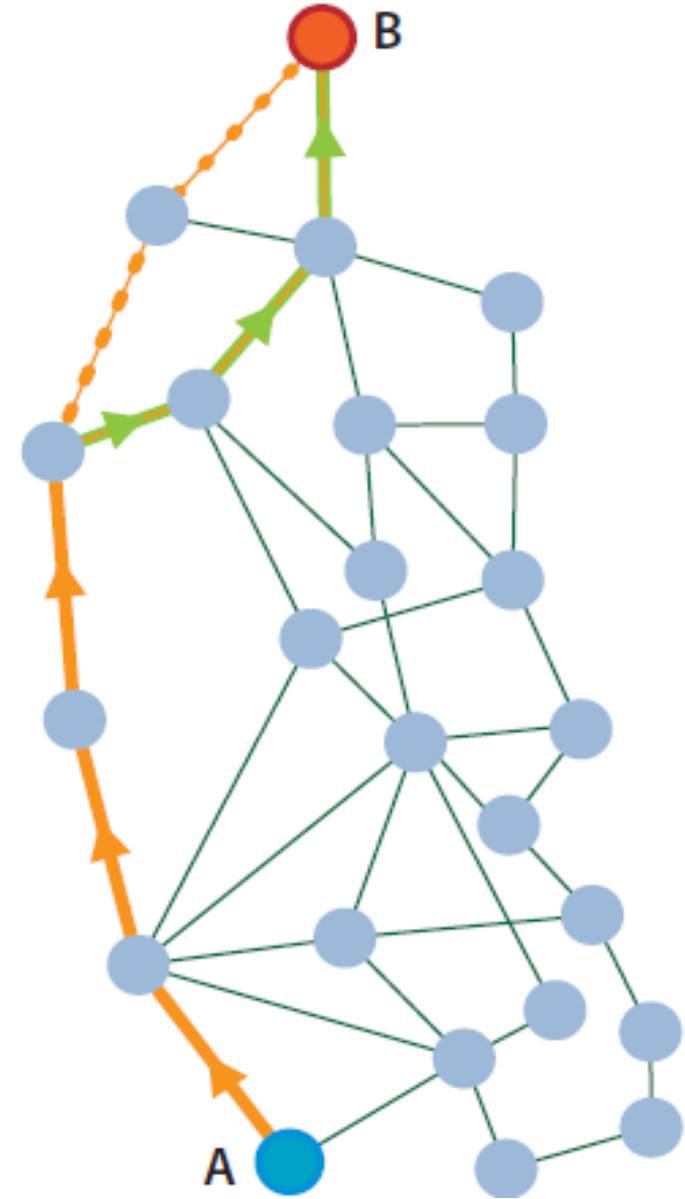
Look for food

- In their search for food, ants wander randomly until they find a source, releasing a certain type of pheromone **[3]**
- What happens in particular is that they create paths with these pheromones to allow others to follow the path to food when found and the path to home as they return
- **Pheromone highways** (trails)

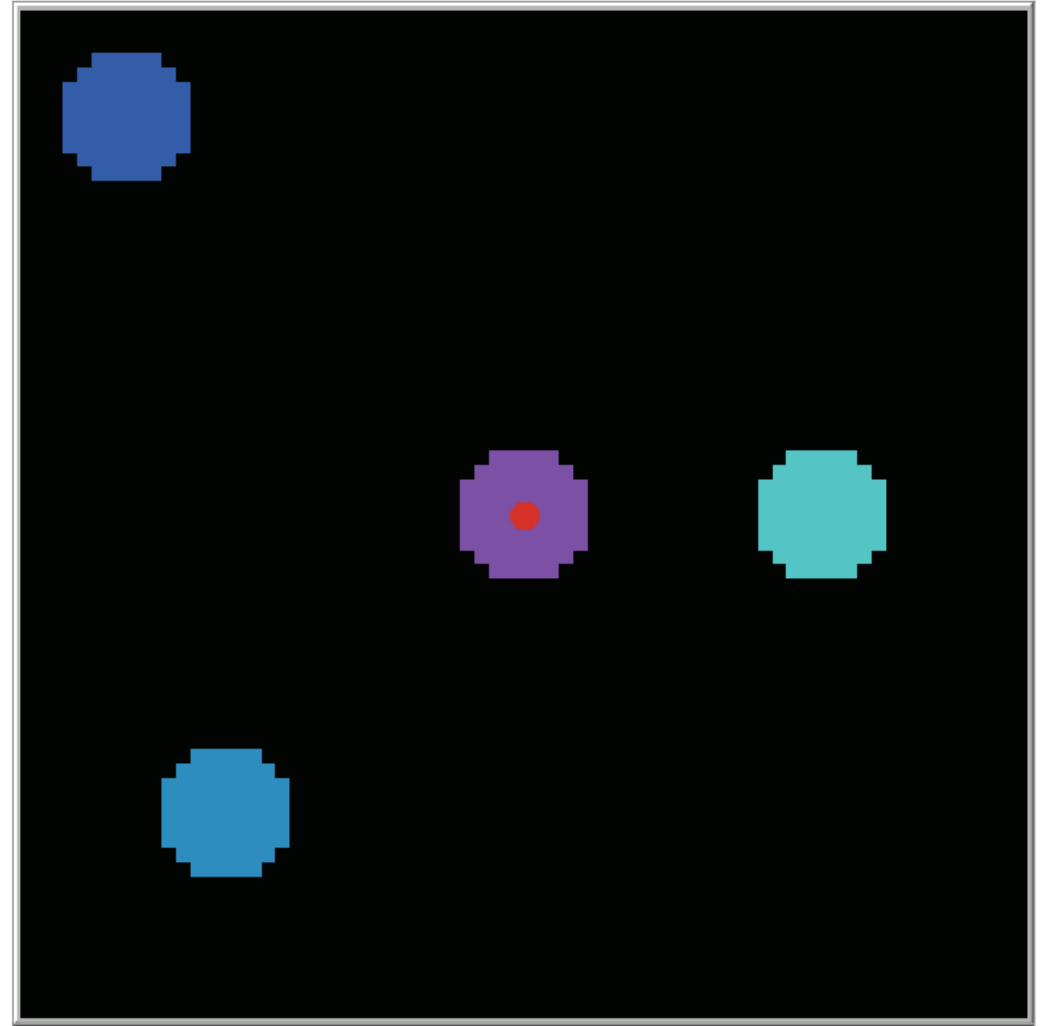


Return to nest

- Ant movement must follow a graph-shaped path
- A single pheromone it's not efficient in simulation
- Pheromones as **nodes** of a graph and ants move creating edges between them

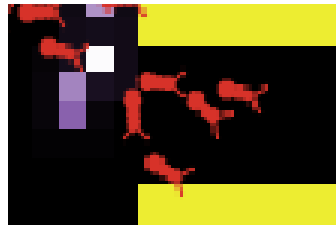


Uri Wilensky's first work and improvements

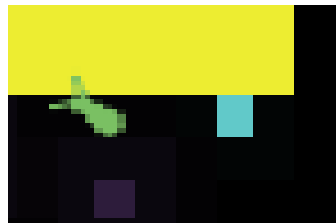


Double pheromone idea

- **Looking-for-food pheromone:** released only while an ant searches for food. This pheromone will help the loaded ants (**colored the same as the food source**) find their way home (**color=purple**).

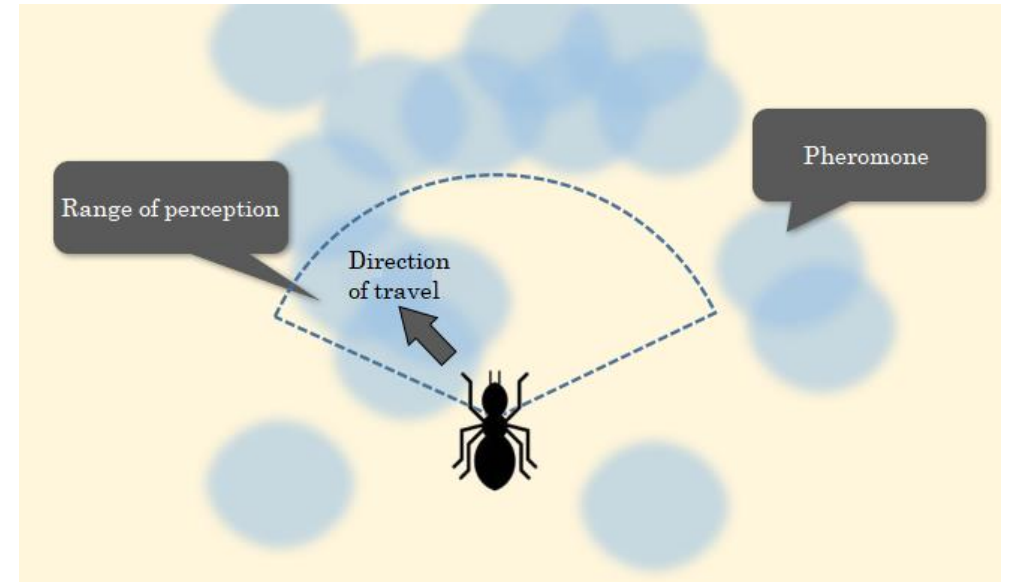


- **Return-to-nest pheromone:** released only when an ant returns to the nest. This pheromone will help the unloaded (**red**) ants find food (**color=cyan**).



Agents

- They are **red** in the search phase and wander randomly until they sense the **food-scent** or **pheromone-food**
- They are **green** or **blue** depending on the source from which they take food as they return to the nest
- They only head for the nest if they are close enough to it (**T_n**)
- They only aim for food if they are close enough to it (**T_f**)
- **T_n <<< T_f**



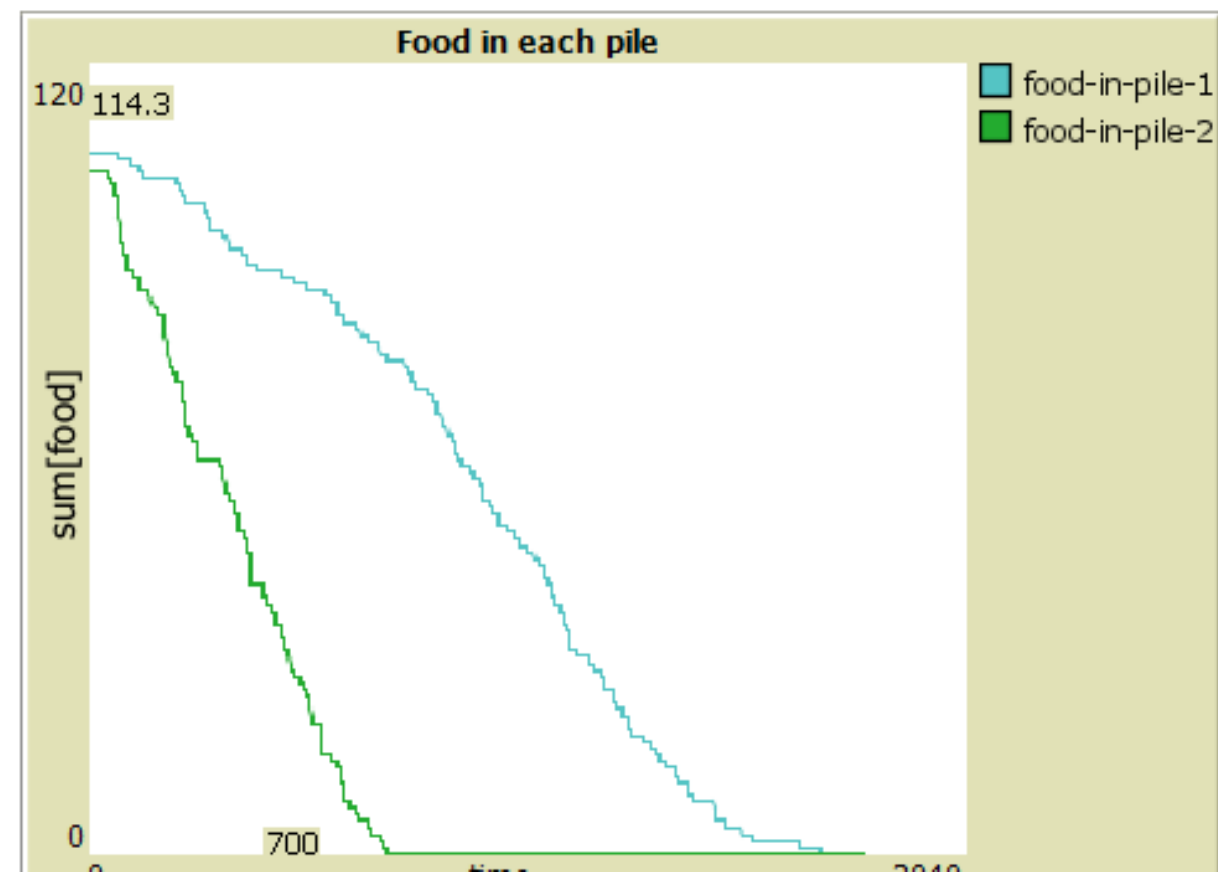
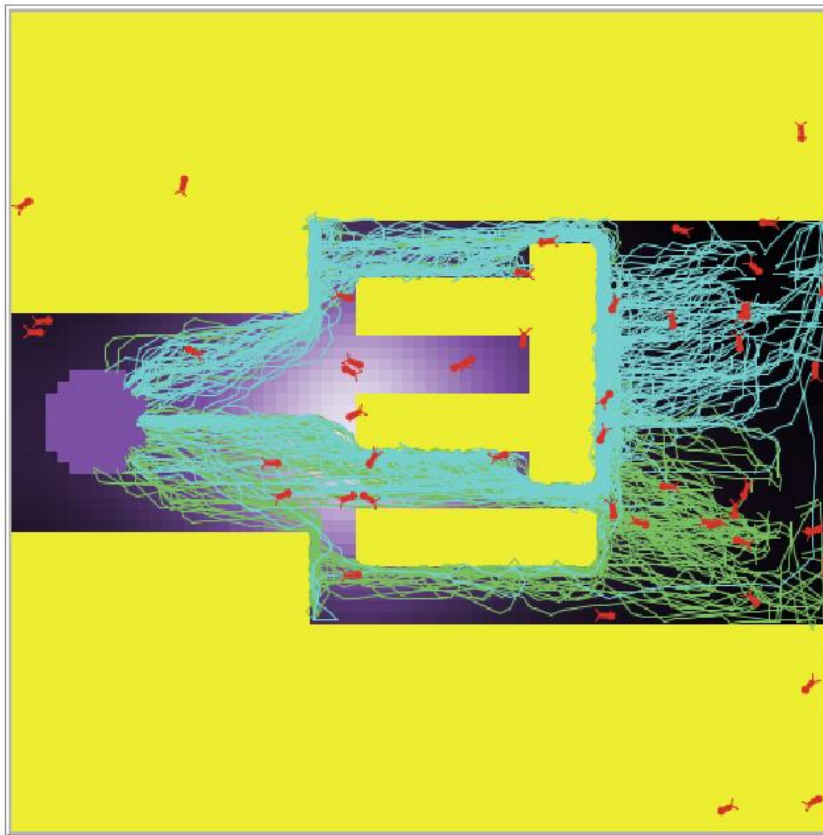
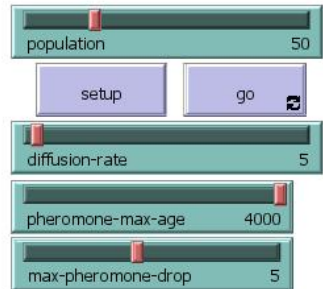
Pheromone aging and lifespan

- To simulate the behavior of pheromones, instead of using an evaporation value (evaporation-rate) which allowed it a very short life, a lifespan value was chosen
- Global variable base ticks (max-pheromone-age)
- The two different pheromones share the same value for max-age
- On average, a pheromone survives for 4,000 ticks and fades very slowly.
- **Water drop effect:** imitate the behavior of a "drop of perfume", which remains in one place for a few moments, and then disappears

Release pheromone and following pheromone

- **Release phase:** pheromone release is controlled by a **counter**. Every 15 steps or so (each ant has its own interval) they release
- Furthermore, there is a counter that limits the release of pheromones by the same ant
- **Following phase:** has a **timer**. To prevent ants from chasing pheromones and circling, they follow the trail and "wait" five seconds before searching for more pheromone. During this time, they continue to move randomly. (**sensing-timer**)

Results with two food sources



Shortest Path

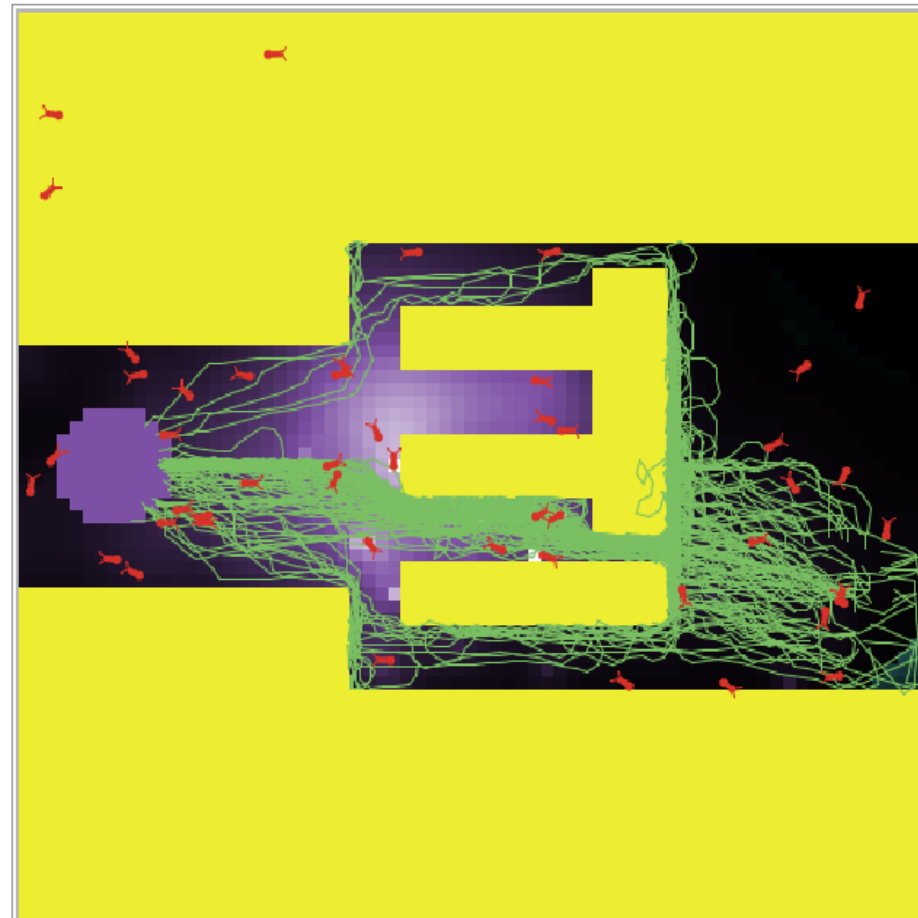
population 50

setup go

diffusion-rate 5

pheromone-max-age 4000

max-pheromone-drop 5



Thanks for the attention!

Daniele Vellani
Nr. Matricola 196186
Distributed AI

References

- [1] H. V. D. Parunak, “‘Go to the Ant’: Engineering Principles from Natural Multi-Agent Systems,” in *Proceedings of the First International Joint Conference on Autonomous Agents and Multiagent Systems: Part 1*, 2002, pp. 1–11. doi: 10.1145/544741.544742.
- [2] Chandrasekhar, A., Gordon, D. M., & Navlakha, S. (2021). A distributed algorithm to maintain and repair the trail networks of arboreal ants. *PLOS Computational Biology*, 17(9), e1009353.
<https://doi.org/10.1371/journal.pcbi.1009353>
- [3] Bonabeau, Eric, and Guy Théraulaz. "Swarm Smarts." *Scientific American*, vol. 282, no. 3, Mar. 2000, pp. 72-79[]